Final Exam Slides

ECE 3011



Georgia Tech College of Engineering
School of Electrical
and Computer Engineering

Updated: Spring 2023

Guidance

- The following 5 slides are what you will use to present for the Final Exam
  - Do not add slides
    - The format of most of these slides are quad-slides
    - This is the expected format in most business communication settings
  - Focus on non-technical communication
  - Consider what your audience would want to see
    - Imagine that your audience is non-technical investors
    - Remember a picture is worth 1000 words

Customer Discovery and Define

- Team Number: **C02**
- Members: **Christopher Semali**

Problem Space

- Customer
  - College students who design/use CAD, Computer Graphics, and or UX/UI software.
- Problem
  - How do you make the navigation and interface controls more intuitive?

Interviews

How many people did you talk to?

- +6

What did you learn?

- That when learning CAD, Computer Graphics, and or UX/UI software there is a steep learning curve and unintuitive design process.

What are the themes?

- Mouse: more precise control, easier to use
- Trackpad: portable (built-into laptops), prone to errors
- 3D motion tracking remotes: (pros) interactive, more intuitive, (cons) lag, latency, in-precise controls

What are the underlying needs?

- To make a steady design process more intuitive from layout to controls, reduce the complexity, and standards across different software's.

Thesis

X will buy Y because of Z

- X – has to be a collection of people (not a hospital)
  - College students
- Y – DO NOT DEFINE THIS YET!
- Z – Why will they buy or use your product/service
  - Provides more intuitive controls ultimately reduces frustrations and design time
  - Promotes gradual learning curve when using a new 3D or UX/UI based software
  - Improves the pipeline between the design and build process

Ideate and Prototype

Thesis

College Students will buy a 3D motion tracking device because it's a tool that can aid their design process and help them learn about perspective drawing, transforming objects, 3D sketching, and constructive design.

Ideate Process

- Talk to people that frequently CAD
- Research different types of Motion Capture Systems
- Discussing the approx. lifespan of the product



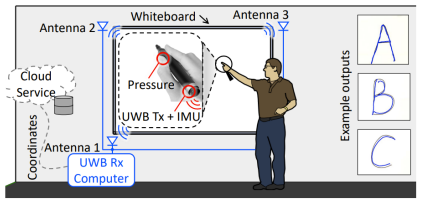


Figure 1: Example ITrackU use for online education.





Hypotheses Statements

- H1. Students have difficulty learning CAD, Computer Graphic, and UX/UI oriented software to be a steep learning curve.
- H2. Students would benefit from using a 3D motion tracking or inertial device as a design tool.
- H3. Students find it important for a device to have quality controls and structred design.

Ugly MVP

1. The End-User holds a pencil with a weight attached to the end of it, while the producer holds a cardboard box object.
2. The end-user will attempt to design a 3D object, by drawing on the box to represent extrusions and a combination of verbal and physical commands to indicate how I should transform/translate the box in 3D space.

Test Results

- Interview sample size: ***8*** | Qualtrics Survey Sample Size: ***25***
- H1. Students have difficulty learning CAD, Computer Graphic, and UX/UI oriented software to be a steep learning curve.
  - 72% of students interviewed have specific difficulties with these software's for a variety of reasons
    - Layout of control functions, tools, keyboard short-cut commands
    - Different set of controls for each software
    - Understanding how to move and transform an object in 2D/3D environment.
    - Hardware constraints

- H2. Students would benefit from using a 3D motion tracking or inertial device as a design tool.
  - Majority of interviewees emphasized the importance of standardizing this devices features across different software platforms, controls, and range of dynamic movement.
  - About 30% of students mentioned it could provide a more fun, intuitive and interactive user experience.

- H3. Students find it important for a device to have quality controls and structured design.
  - About 80% of students provided advantages of mouse and disadvantages of track pad.
  - The responses in favor of a mouse, primarily mentioned that it has better controls
  - The responses in favor of a trackpad, emphasized it’s compact and portable.
- <u>Side note:</u>
  - Many students' problems surrounding CAD software are mostly associated with trying to find a unique feature that enable them to transform the object. Mitigating the time spent search for the location of a feature is what interests them more than new hardware.

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CREATING THE NEXT

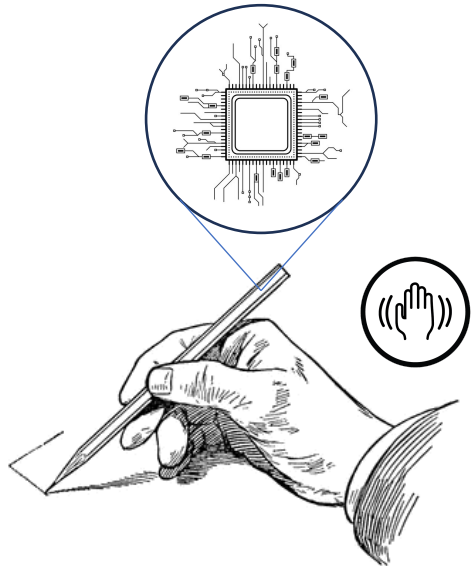
Final Idea Pitch – One slide project summary of MVP

Background

• In a world, where designing products is overly complicated, frustrating, and time-consuming for new learners! We seek to create a device that simplifies and speeds up the design process. Since most professional engineers and designer received their original training in college, our target audience is college students newly learning Computer Aided Design (CAD), Computer Graphics, and UX/UI software.

Solution

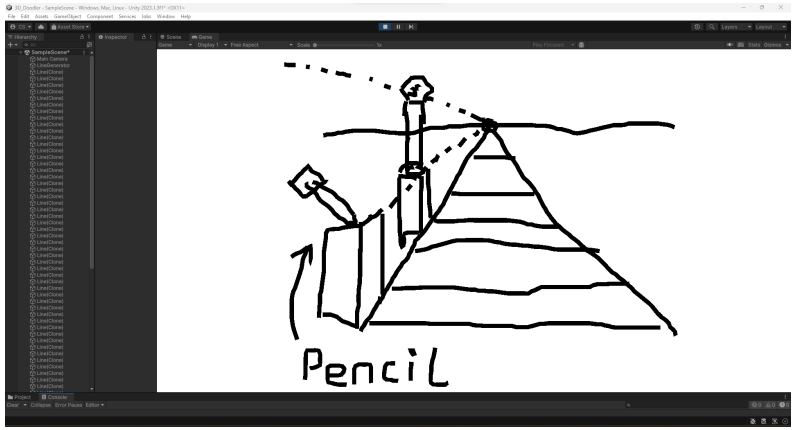
- An electronic attachment to a pencil, which can record data that tracks it movement across a flat surface and in 3D space.
- This device will be especially helpful when learning/applying concepts such as perspective drawing, transforming objects, 3D sketching, and constructive design.



Testing and Market Survey

- A motion tracking device that can track and record movement information from a 3D environment. (Tested within 2D space)
- Students would prefer having electronic devices with long battery life, sleek design, and comfortable to hold for an extended period
  - weighting under 0.11 pounds (mass of 50 grams).
- College Students would prefer more intuitive controls whenever manipulating 3D objects in software

MVP Picture



Unity Game: 2D Doodler

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Conclusion

Go/No Go Decision

Go!

- This could be a very resourceful educational tool, whose software can be adaptable to various real-world applications.
- The hypotheses were true.

Surprises

The specific, yet pain points students experience while using CAD software's was eye-opening and goes to show that every person/end-user has different set of needs.

Class Feedback

- Make it 9:30 class
- Exercises to practice using Git & GitHub, similar to CS 2340 Objects & Design